

Chapter 5, Section 3

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August 7, 2026

1. Let X be a nonsingular projective variety of any dimension, let Y be a nonsingular subvariety, and let $\pi: \tilde{X} \rightarrow X$ be obtained by blowing up Y . Show that $p_a(\tilde{X}) = p_a(X)$.

Proof. We can generalize the proof of (3.4).

Proposition 0.1. *In the above setting, we have $\pi_* \mathcal{O}_{\tilde{X}} = \mathcal{O}_X$, and $R^i \pi_* \mathcal{O}_{\tilde{X}} = 0$ for $i > 0$. Therefore, $H^i(X, \mathcal{O}_X) \cong H^i(\tilde{X}, \mathcal{O}_{\tilde{X}})$ for all $i \geq 0$.*

By (II, 8.24), the strict transform of Y in $\tilde{X}$ is isomorphic to the projection space bundle $\mathbb{P}(\mathcal{I}/\mathcal{I}^2)$, where $\mathcal{I}$ is the ideal sheaf $\mathfrak{f}$. It is locally free of rank $r = \text{codim}(Y, X)$ since Y is nonsingular. In particular, $\mathbb{P}(\mathcal{I}/\mathcal{I}^2)$ has dimension $Y + \text{codim } r - 1$, so it corresponds to a divisor on X . By (II, 7.15), there exists a commutative diagram

$$\begin{array}{ccc} \tilde{Y} & \xrightarrow{\tilde{f}} & \tilde{X} \\ \downarrow & & \downarrow \\ Y & \xrightarrow{f} & X \end{array}$$

where f is a closed immersion, and so is $\tilde{f}$. By the theorem on formal functionns, For any scheme theoretic point $y \in Y$, the natural map

$$\hat{\mathcal{F}}^i_y = (R^i \pi_* \mathcal{O}_{\tilde{X}})_y \otimes \hat{\mathcal{O}}_{y, Y} \cong \varprojlim H^i(Y_n, \mathcal{O}_{Y_n}).$$

By (7.11), $\pi_* \mathcal{O}_{\mathbb{P}(\mathcal{I}/\mathcal{I}^2)} = \mathcal{O}_X$

Since π is an isomorphism of $V = \tilde{X} - E$ onto $U = X - Y$ (II, 7.13). The sheaves $\mathcal{F}^i = R^i \pi_* \mathcal{O}_{\tilde{X}}$ for $i > 0$ have support at Y . $\square$

2. Let C and D be curves on a surface X , meeting at a point P . Let $\pi: \tilde{X} \rightarrow X$ be the monoidal transformation with center P . Show that $\tilde{C} \cdot \tilde{D} = C \cdot D - \mu_P(C) \cdot \mu_P(D)$. Conclude that $C \cdot D = \sum \mu_P(C) \cdot \mu_P(D)$, where the sum is taken over all intersection points of C and D , including infinitely near intersection points.
3. Let $\pi: \tilde{X} \rightarrow X$ be a monoidal transformation, and let D be a very ample divisor on X . Show that $2\pi^*D - E$ is ample on $\tilde{X}$.
4. *Multiplicity of a Local Ring.* Let A be a Noetherian local ring with maximal ideal $\mathfrak{m}$. For any $l > 0$, let $\psi(l) = \text{length}(A/\mathfrak{m}^l)$. We call ψ the *Hilbert-Samuel function* of A .
 - (a) Show that there is a polynomial $P_A(z) \in \mathbb{Q}[z]$ such that $P_A(l) = \psi(l)$ for all $l \gg 0$. This is the *Hilbert-Samuel polynomial* of A .
 - (b) Show that $\deg P_A = \dim A$.
 - (c) Let $n = \dim A$. Then we define the *multiplicity* of A , denoted $\mu(A)$, to be $(n!)$ · leading coefficient of P_A . If P is a point on a Noetherian scheme X , we define the *multiplicity* of P on X , $\mu_P(X)$, to be $\mu(\mathcal{O}_{P, X})$.
 - (d) Show that for a point P on a curve C on a surface X , this definition of $\mu_P(C)$ coincides with the one in the text just before (3.5.2).
 - (e) If Y is a variety of degree d in $\mathbb{P}^n$, show that the vertex of the cone over Y is a point of multiplicity d .

5. Let $a_1, \dots, a_r$, $r \geq 5$, be distinct elements of k , and let C be the curve in $\mathbb{P}^2$ given by the (affine) equation $y^2 = \prod_{i=1}^r (x - a_i)$. Show that the point P at infinity on the y -axis is a singular point. Compute δ_P and $g(\tilde{Y})$, where $\tilde{Y}$ is the normalization of Y . Show in this way that one obtains hyperelliptic curves of even genus $g \geq 2$.
6. Show that analytically isomorphic curve singularities (I, 5.6.1) are equivalent in the sense of (3.9.4), but not conversely.
7. For each of the following singularities at $(0,0)$ in the plane, give an embedded resolution, compute δ_P , and decide which ones are equivalent.
- (a) $x^3 + y^5 = 0$
 - (b) $x^3 + x^4 + y^5 = 0$
 - (c) $x^3 + y^4 + y^5 = 0$
 - (d) $x^3 + y^5 + y^6 = 0$
 - (e) $x^3 + xy^3 + y^5 = 0$

Proof. For each $f(x, y) = 0$, denote $F(x, y, z) = 0$ the projectivization. For each homogenous degree r polynomial $F(x, y, z) \in k[x, y, z]$, we compute the singular locus of F as the ideal $J = (F, \partial_x F, \partial_y F, \partial_z F)$, where

$$\partial_x F := \frac{\partial F}{\partial x}, \quad \partial_y F := \frac{\partial F}{\partial y}, \quad \partial_z F := \frac{\partial F}{\partial z}$$

Notice that $\mu_P(f) = 3$ for all f above. In particular, $\delta_P = 3$.

- (a) The curve is parametrized by

$$z \mapsto (z^5, -z^3).$$

The tangent vector vanishes at an order of 2, which means $\mu_P(C) = 3$.

Blowing up in coordinates $x = ty$ gives

$$y^3(t^3 + y^2) = 0,$$

where the exceptional divisor E_1 is defined by $x = y = 0$. Then $\pi^*C = \tilde{C} + 3E$. Since $\tilde{C}$ is still singular, parametrized by

$$z \mapsto (x, y, t) = (iz^5, iz^3, z^2).$$

The tangent vector vanishes at an order of 1, so we have $\mu_P(\tilde{C}) = 2$, so $\delta_P = 1$. It is either the cusp or node,

- (b)
- (c)
- (d) $t^3 + y^2 + y^3 = 0$, cusp
- (e) $t^3 + ty + y^2 = 0$, node

□

8. Show that the following two singularities have the same multiplicity, and the same configuration of infinitely near singular points with the same multiplicities, hence the same δ_P , but are not equivalent.

- (a) $x^4 - xy^4 = 0$
- (b) $x^4 - x^2y^3 - x^2y^5 + y^8 = 0$